

Computational Engineering

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Assignment 3: Burgers Equation

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1 Problem specification

In this work, the one-dimensional Burgers' equation is solved.

The governing equation is:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2} \tag{1}$$

where $u(x,t)$ represents the velocity field and $\nu = 1/Re$ is the kinematic viscosity.
The main objective is to analyze fundamental aspects of turbulence, such as the energy cascade, the interaction between scales, and the role of the convective and diffusive terms. Two main configurations are proposed:

- $Re = 40$ with $N = 20$ Fourier modes with the aim of observing an underresolution of the model.
- $Re = 40$ with higher Fourier modes with the aim of achieving a DNS resolution.

First, the problem is solved without a turbulence model (Direct Numerical Simulation, DNS), and in a second stage, a Large Eddy Simulation (LES) model is implemented by introducing a spectral eddy viscosity.

2 Theoretical framework

2.1 Burgers in Fourier space

To facilitate the calculation of the convective and diffusive terms (derivatives), to naturally analyze the evolution of the energy in each mode, and to facilitate the study of interscale interactions, the numerical resolution of the Burgers equation in Fourier space is proposed.

Applying this transformation, the Burgers equation in Fourier becomes:

$$\frac{d\hat{u}_k}{dt} + \sum_{k=p+q} \hat{u}_p(iq)\hat{u}_q = -\nu k^2 \hat{u}_k \tag{2}$$

The first term represents the temporal evolution of each mode, the second corresponds to the nonlinear interaction between modes (convolution), and the third is the viscous dissipation.

The velocity function $u(x,t)$ can be reconstructed from its Fourier coefficients through:

$$u(x,t) = \sum_{k=-N}^{N} \hat{u}_k(t)e^{ikx} \tag{3}$$

2.2 Numerical treatment

To the equation numerically, an explicit Euler scheme is used to discretize the temporal derivative.

The time step Δt is determined by a CFL-type condition to ensure the numerical stability of the scheme:

$$\Delta t = \text{CFL} \times \frac{1}{\nu N^2}, \tag{4}$$

where N is the maximum resolved wavenumber.

Since the physical velocity field $u(x,t)$ must remain real, the Fourier coefficients are constrained at each time step to satisfy:

$$\hat{u}_k = \overline{\hat{u}_{-k}}. \tag{5}$$

3 Turbulence model: LES in Fourier space

When solving the Burgers equation with few scales (few Fourier modes), DNS is not sufficiently accurate. To improve the solution, LES model is applied.

The LES model starts by filtering the Burgers equation:

$$\partial_t \bar{u} + \bar{u}\partial_x \bar{u} = \frac{1}{Re}\partial_{xx}\bar{u} + f - \partial_x \tau(u) \tag{6}$$

where $\tau(u)$ is the *subfilter tensor*, defined and aproximated in the following way.

$$\tau(u) = \overline{u^2} - \bar{u}^2 \approx \nu_t \partial_x \bar{u} \tag{7}$$

where the *turbulent viscosity* ν_t is modeled as:

$$\nu_t\left(\frac{k}{k_N}\right) = \nu_t^{\infty}\left(\frac{E_{k_N}}{k_N}\right)^{1/2} \nu_t^*\left(\frac{k}{k_N}\right) \tag{8}$$

where:

$$\nu_t^{\infty} = 0.31\frac{5-m}{m+1}\sqrt{3-m}C_K^{-3/2} \tag{9}$$

$$\nu_t^*\left(\frac{k}{k_N}\right) = 1 + 34.5\, e^{-3.03\left(\frac{k_N}{k}\right)} \tag{10}$$

Finally, the practical change in the code consists of modifying the effective viscosity as:

$$\nu_{\text{eff}}(k) = \nu + \nu_t(k) \tag{11}$$

thus allowing to correctly simulate the unresolved scales using LES.

4 Code Structure

The code developed for the resolution of the Burgers equation in Fourier space is organized into the following main blocks:

- **Definition of parameters:** The values for the number of modes N, Reynolds number Re, kinematic viscosity ν, the Courant–Friedrichs–Lewy number (CFL), the maximum simulation time $t_{\max}$, and the specific parameters of the LES model (coefficient C_K and exponent m) are set.
- **Initialization of variables:**
  - The vector of modes k in the interval $[-N, N]$ is defined.
  - The Fourier coefficient vectors u_k are initialized for both the DNS and the LES simulations.
  - The condition $u_0 = 0$ and the conjugate symmetry $u_{-k} = \overline{u_k}$ are imposed.
- **Calculation of turbulent viscosity (only for LES):** A turbulent viscosity dependent on k is computed based on the proposed spectral model, proportional to $\left(\frac{|k|}{N}\right)^m$.
- **Time loop:** The temporal evolution of the Fourier coefficients is solved over n_t time steps using an explicit Euler scheme:
  - The convective term (C) for each mode k is calculated using triadic interactions.
  - The diffusive term (D) for DNS and LES is calculated, including the turbulent viscosity in the case of LES.
  - The values of u_k are updated for DNS and LES.
  - The first mode u_1 is forced to remain constant throughout the simulation.
- **Calculation of the energy spectrum:** At the end of the simulation, the energy spectrum E_k is computed for the positive modes.
- **Graphical representation:** The energy spectra are plotted in log-log scale for each value of N, comparing the solutions obtained through DNS and LES. Additionally, the energy as a function of time is plotted for each of the specified values of N.

5 Code verification

The result obtained from the code is the following:

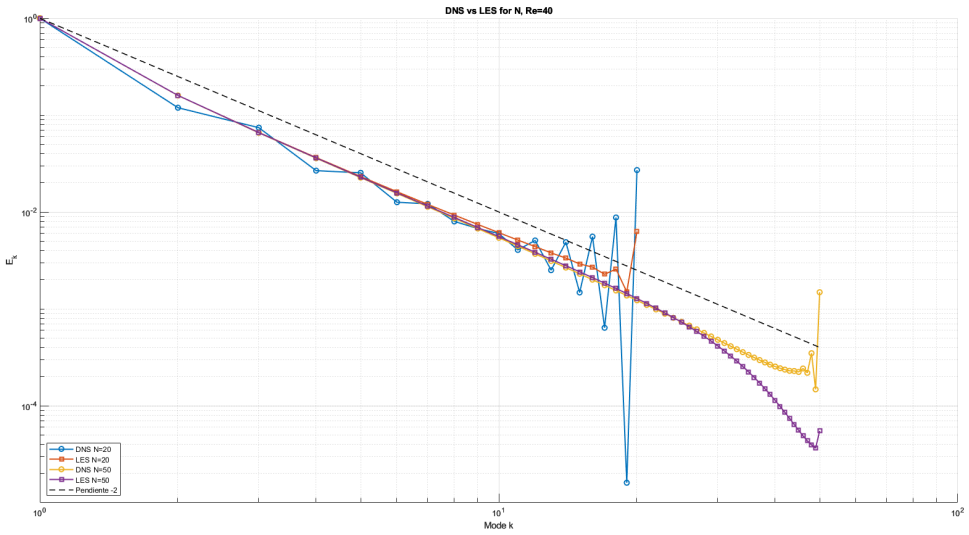


Figure 1: Turbulent kinetic energy as a function of modes. N=[20, 50]. t=5s, C=0.01. LES: C_K=0.05; m=2

This figure shows the energy spectrum E_k as a function of the mode k for **DNS** and **LES** simulations at $Re = 40$, comparing resolutions $N = 20$ and $N = 50$.

- For low modes (small k), DNS and LES match well, since large scales are properly resolved.
- For high modes (large k), DNS with $N = 20$ shows dispersion due to lack of resolution, while LES stabilizes the solution by applying turbulent viscosity.
- Increasing N to 50 improves DNS accuracy throughout the spectrum, approaching the theoretical slope of -2.

5.1 Behavior at $Re = \infty$

When the Reynolds number tends to infinity, the effects of viscous diffusion become negligible compared to convective effects. In this regime, energy transfer between scales occurs mainly through nonlinear interactions (convection), without significant energy loss due to dissipation.
As a result, the system reaches a self-similar state in which the energy distribution among the modes follows a power law. In particular, for the one-dimensional Burgers equation, it is demonstrated that the energy spectrum E_k decays with the wavenumber as:

$$E_k \sim k^{-2}. \tag{12}$$

This implies that, in a logarithmic (log-log) representation, the energy spectrum appears as a straight line with slope -2. This behavior can be observed in the simulations when Re is sufficiently large and the flow reaches a statistically stationary state.
As can be observed, the code fulfills this premise:

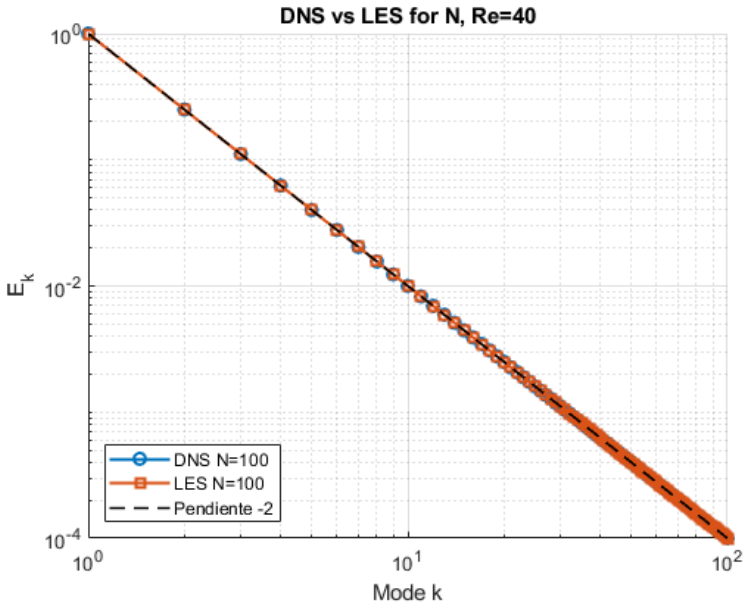


Figure 2: Turbulent kinetic energy (N=100) for $\nu = 0$

5.2 Evolution of total energy over time

During the simulation, the temporal evolution of the total energy $E(t)$ is analyzed for different spatial resolutions, specifically for $N = 20$, $N = 30$, and $N = 40$. As observed in the graph, in all cases the energy progressively decreases over time, which is expected behavior due to the action of viscosity dissipating energy at small scales.

Initially, there is a rapid drop in energy. This is due to the strong energy transfer between different modes. During this transient phase, the modes redistribute until reaching a more stable configuration, where energy is transferred from low-frequency modes to high-frequency modes, where it is finally dissipated.

As the number of modes N increases, the resolution of the simulation improves, allowing a more accurate capture of energy transfer processes towards smaller scales. Consequently, the oscillations appearing in the energy curve for lower values of N are reduced.

Finally, in all cases, the total energy tends to stabilize as time progresses, indicating that the system approaches a stationary state.

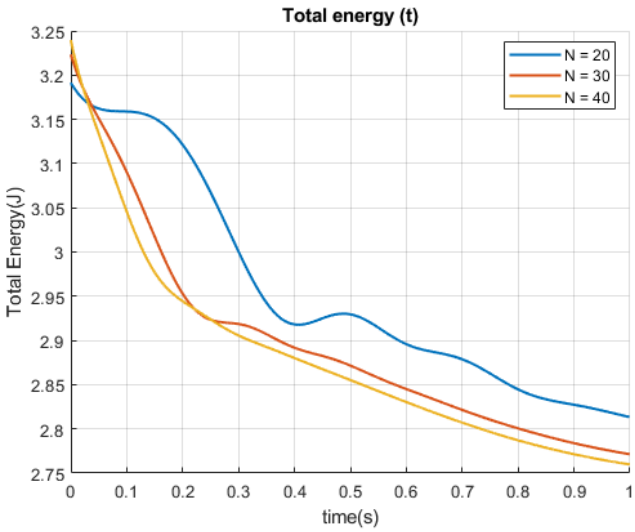


Figure 3: Energy as a function of time for N=[20, 30, 40]. Re=40. CFL coefficient=0.01. DNS

6 Numerical Studies

6.1 Mode error

The objective of this study is to understand how the energy balance error evolves for each mode k and for different values of N.

The theoretical equation for the transport of the kinetic energy of each mode is given by:

$$Error = \left| -\frac{dE_k}{dt} - \frac{2k^2}{Re}E_k - \text{Convection} \right|, \tag{13}$$

where E_k represents the energy of mode k, $\frac{2k^2}{Re}E_k$ corresponds to the loss of energy due to viscous diffusion, and the convection term groups the nonlinear interactions between modes.

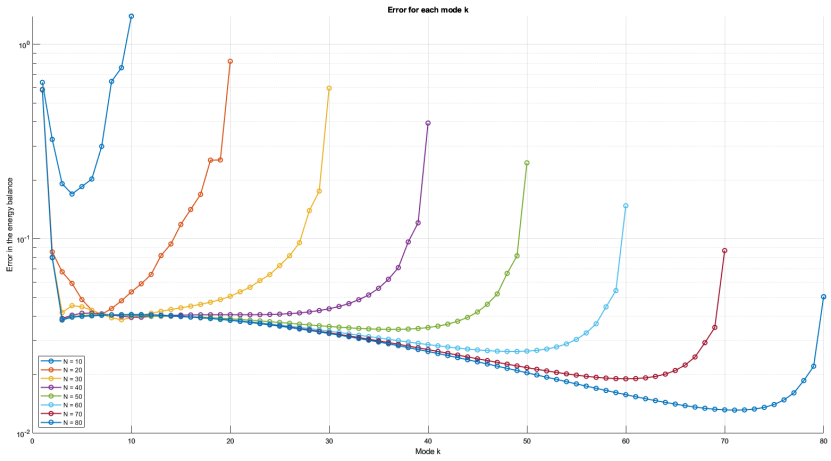


Figure 4: Mean energy balance error for each mode k and different values of N.

It is observed that:

- For intermediate modes, the error is small, indicating that the energy balance is adequately satisfied.
- In very high-frequency modes (close to N), the error grows significantly due to the loss of numerical precision and diffusion effects.
- As N increases, the mean error decreases, reflecting a better spectral resolution of the problem.

This confirms that our numerical scheme correctly reproduces the dynamics of the Burgers equation in Fourier space.

6.2 Influence of the Reynolds number on the energy spectrum

To evaluate how the cascade of turbulent kinetic energy evolves for different Reynolds numbers, a comparative graph has been made:

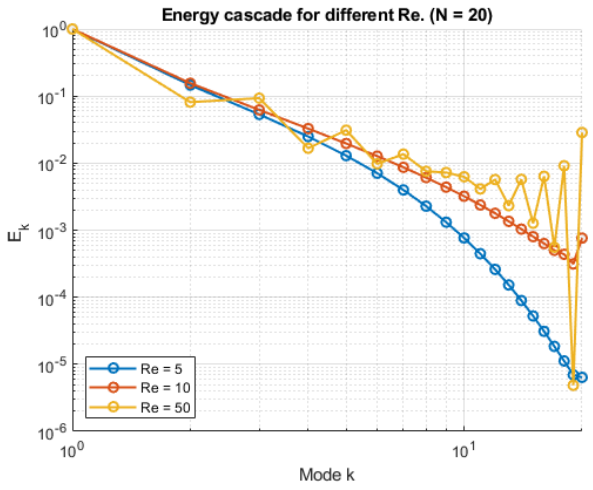


Figure 5: Turbulent kinetic energy as a function of the modes k for different Reynolds numbers. DNS

- In low-frequency modes (small k), the energy is similar for different Re, since viscous diffusion hardly affects the large scales.
- In high-frequency modes (large k), the effect of Re is more noticeable. For low Re, energy dissipates quickly with k. For high Re, energy persists in higher-frequency modes before being dissipated.
- In the intermediate range, energy follows a power law. For the Burgers equation, a slope of approximately k^{-2} is expected.
- At very high frequencies, oscillations may appear due to numerical errors.

7 Conclusion

In this work, the Burgers equation has been implemented and solved in Fourier space, using both direct numerical simulation (DNS) and a large eddy simulation (LES) model.

Additional validation studies have been carried out, such as the evolution of the total energy as a function of time, the evolution of the system with the variation of the Reynolds number and the error of the energy balance, which have confirmed the correct performance of the numerical scheme. This work has deepened the understanding of fundamental turbulence concepts, such as energy transfer between scales and the role of viscosity.

Although the code developed successfully solves the problem and captures the expected physical behavior, the computation time has proven to be relatively long, especially when increasing the number of modes N. This suggests that the algorithm could be further optimized, for example by improving the way triadic interactions are computed or by implementing more efficient numerical schemes.